

Photoacoustic Tomography Image Reconstruction

Simultaneous Reconstruction of Light Absorption and Sound Speed Fields Using Multiple Illumination Angles and Learned Regularization in a Limited View Setting

Matthew Grindstaff

University College London

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I, Matthew Grindstaff, confirm that the work presented in this dissertation is my own. Where information has been derived from other sources, I confirm that this has been indicated in the dissertation.

Abstract

A practical approach to simultaneously reconstructing the light absorption and sound speed fields in photoacoustic tomography (PAT) is presented, leveraging multiple illumination angles and a learned regularization method. While the reconstruction task is simplified under the assumption of a homogeneous sound speed field, real biological tissues exhibit significant heterogeneity, leading to distortions in these reconstructions and the ill-posed problem of reconstructing the sound speed field. The use of multiple illumination angles is shown to enhance the uniqueness of the sound speed field solution, reduce "shadows", and improve the parallelism of the reconstruction algorithm. Incorporating a convolutional neural network (CNN) to learn gradient updates for the sound speed field, structural and dynamic similarities between these fields can be utilized to recover features on different scales.

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The source code for this project can be found here: <https://github.com/grindstm/pat.git>

1 Introduction

1.1 Photoacoustic Tomography

In Photoacoustic Tomography (PAT), a nano-second pulse of laser light illuminates the interior of biological tissue. Chromophores in the tissue absorb the light and exhibit a thermoelastic pressure spike. These waves propagate through the tissue and are detected by ultrasound transducers. The transducer pressure data is then reconstructed to form an image of the initial pressure spike p_0 which is coincident with the tissue geometry. Photoacoustic imaging poses a range of advantages over other modalities. The laser pulses are assumed to be shorter than the stress and thermal confinements of the illuminated material, potentiating image reconstructions with higher spatial resolution and contrast compared to ultrasound imaging [11]. High-frequency acoustic waves are required to carry high spatial frequency information associated with small structures in the subject. These frequencies are attenuated through absorption and scattering more rapidly than lower frequencies during propagation [8], limiting the attainable image resolution by depth. Still, acoustic waves scatter about 1000 times less than photons, so depths ranging from single millimeters up to about 7 centimeters are feasible for in vivo imaging [9].

In a medium with homogeneous sound speed, the solution to the inverse problem of reconstructing the initial pressure field is unique, provided that the detector configuration is sufficient to capture all the necessary information about the pressure waves emanating from their source. This typically requires dense placement of detectors around the subject to record the complete set of wavefronts [1]. In ideal cases, such as with spherical or cylindrical geometries, the wave equation has explicit analytical solutions. For example, in a spherical geometry, the spherical Radon transform can be explicitly inverted to reconstruct the initial pressure field. However, this method requires precise detector placement, with transducers ideally positioned on a spherical surface surrounding the subject. Better measurements can be acquired when transducers maintain direct contact with the subject, often necessitating the use of ultrasound gel. This prevents acoustic waves from encountering the highly attenuating skin-air impedance boundary. While the spherical or semi-spherical sensor arrangement eases the task of image reconstruction, it is often impractical due to the difficulty of maintaining perfect contact and coverage around the entire subject. Given these challenges, there is a need for reconstruction methods that can handle limited-view configurations and generally, arbitrary sensor placement.

A heterogenous sound speed field significantly complicates the reconstruction problem. Scattering and refraction disturbs the wave field, requiring computationally demanding discretized solutions to the forward model [2]. Without accurate knowledge of the sound speed field c , reconstructions of p_0 can be blurry or distorted. Biological material has a complex, spatially varying sound speed, with fat having a speed of sound of about 1440m/s and blood having a speed of sound of about 1540m/s.

Recovering both the initial pressure field, p_0 , and the sound speed field, c , from measurements taken during a single illumination is comparable to reconstructing a 3D model from a single photograph. The challenge in this problem stems from the inherent ambiguity in determining the sound speed field. Specifically, a given set of transducer data, generated by an unknown p_0 distribution, can be attributed to multiple different sound speed fields. If this ambiguity is not addressed, the reconstructed sound speed may converge to an arbitrary solution that merely minimizes the data loss,

rather than accurately representing the true field.

The ill-posed nature of this problem is exacerbated by the non-linear coupling of the sound speed and initial pressure fields in the wave equation. However, in this case, the fields are assumed to exhibit a direct structural similarity, which is crucial to exploit during the reconstruction process. The reconstruction method employed here involves repeatedly simulating the wave equation, a process that is computationally intensive. Therefore, it is essential to strategically address the ill-posedness to ensure that the reconstruction is both accurate and efficient.

One strategy which can improve the uniqueness of the sound speed field solution is to illuminate the subject from multiple angles. This results in a set of measurements that differ in structure and provide enough information that the optimization process can converge to a more accurate solution.

Finally, a convolutional neural network (CNN) is used to effectively share information between the fields by learning a gradient update for the sound speed field which incorporates the absorption field, previous sound speed field, and the gradients of these fields obtained from backpropagating the data residual.

Note that all 2D images in the original pdf version of this document are uncompressed, so zoom in to view details.

2 Formulation

Forward model: This describes the propagation of waves in a medium. The acoustic wave equation expresses this:

$$\frac{\partial^2 p(\mathbf{x}, t)}{\partial t^2} - c(\mathbf{x})^2 \nabla^2 p(\mathbf{x}, t) = 0 \quad \frac{\partial p(\mathbf{x}, 0)}{\partial t} = 0 \quad (1)$$

where $p(\mathbf{x}, t)$ is the pressure at position $\mathbf{x}$ and time t , $c(\mathbf{x})$ is the spatially varying sound speed.

Illumination Model: This simplified illumination model relates the absorption field to the initial pressure field, neglecting scattering and refraction. Assuming light intensity decays exponentially with the depth that it travels into material due to absorption and that it is projected from a specific angle, the initial pressure p_0 can be modelled as:

$$p_0(\mathbf{x}, \phi) = \mu(\mathbf{x}) I_0 e^{-\int_0^{d(\mathbf{x}, \phi)} \mu(\mathbf{x}') ds} \quad (2)$$

where $\mu(\mathbf{x})$ is the absorption coefficient, I_0 is the initial light intensity, $d(\mathbf{x}, \phi)$ is the depth within the material, dependent on the angle of illumination ϕ . The integral $\int_0^{d(\mathbf{x}, \phi)} \mu(\mathbf{x}') ds$ represents the accumulated absorption along the path of the light as it travels through the material from the surface to the point x . For ease, a significantly simplified model is used at this stage which neglects the path integral:

$$p_0(\mathbf{x}, \phi) = \mu(\mathbf{x}) I_0 e^{-\alpha d(\mathbf{x}, \phi)} \quad (3)$$

where α is a constant attenuation coefficient. This simplification allows a clean separation between the absorption field and what has been termed an attenuation mask, $\mathcal{U}$.

Forward Operator $\mathcal{A}$: The solution to (1) can be written:

$$\mathbf{p}_{\text{data}} = \mathcal{A}(\mathbf{p}_0, \mathbf{c}) \quad (4)$$

where:

- $\mathbf{p}_{\text{data}}$ is an array of time-series of pressures at ultrasound transducer in the domain.
- $\mathbf{p}_0 = \mathcal{U}(\phi)$ follows from Equation 3 and is an initial pressure distribution as a function of the absorption field and illumination angle.

Optimisation problem The aim is to simultaneously find a $\boldsymbol{\mu}$ and $\mathbf{c}$ satisfying (3) by minimizing the mean squared error (MSE) between the data and the simulation applied to the initial pressure field:

$$(\boldsymbol{\mu}^*, \mathbf{c}^*) = \arg \min_{\boldsymbol{\mu}, \mathbf{c}} \{ \|\mathbf{P}_{\text{data}} - \mathcal{A}(\mathbf{P}_0(\boldsymbol{\mu}, \phi), \mathbf{c})\|_2^2 \} \quad (5)$$

Multi-illumination Applying the illumination model in Equation 3 at various angles gives a set of initial pressure fields:

$$\mathbf{P}_0(\boldsymbol{\mu}, \phi) = [\mathbf{p}_0(\boldsymbol{\mu}, \phi_1), \mathbf{p}_0(\boldsymbol{\mu}, \phi_2), \dots, \mathbf{p}_0(\boldsymbol{\mu}, \phi_K)] \quad (6)$$

for illumination angles $\phi = 1, \dots, \Phi$, the measurements $\mathbf{p}_{\text{data}}^{(\phi)}$ corresponds to illumination angles ϕ . Let $\mathbf{P}_{\text{data}} = [\mathbf{p}_{\text{data}}^{(1)}, \mathbf{p}_{\text{data}}^{(2)}, \dots, \mathbf{p}_{\text{data}}^{(\Phi)}]$ be the combined measurement data for the set of illuminations. Finding a single $\boldsymbol{\mu}$ and a single $\mathbf{c}$ remains the objective and the optimization problem is reformulated:

$$\mathcal{L}_{\text{rec}} = \|\mathbf{P}_{\text{data}} - \mathcal{A}(\mathbf{P}_0, \mathbf{c})\|_2^2 \quad (7)$$

$$(\boldsymbol{\mu}^*, \mathbf{c}^*) = \arg \min_{\boldsymbol{\mu}, \mathbf{c}} \mathcal{L}_{\text{rec}} \quad (8)$$

where $\mathcal{L}_{\text{rec}}$ is the reconstruction loss.

Solution A gradient descent approach:

$$\nabla_{\boldsymbol{\mu}} \mathcal{L}_{\text{rec}} = \nabla_{\boldsymbol{\mu}} \|\mathbf{P}_{\text{data}} - \mathcal{A}(\mathbf{P}_0(\boldsymbol{\mu}, \phi), \mathbf{c})\|_2^2 \quad (9)$$

$$\nabla_{\mathbf{c}} \mathcal{L}_{\text{rec}} = \nabla_{\mathbf{c}} \|\mathbf{P}_{\text{data}} - \mathcal{A}(\mathbf{P}_0, \mathbf{c})\|_2^2 + \mathcal{R}_{\mathbf{c}}(\boldsymbol{\mu}, \nabla_{\boldsymbol{\mu}} \mathcal{L}_{\text{rec}}, \mathbf{c}, \nabla_{\mathbf{c}} \mathcal{L}_{\text{rec}}) \quad (10)$$

where $\mathcal{R}_{\mathbf{c}}(\boldsymbol{\mu}, \nabla_{\boldsymbol{\mu}} \mathcal{L}_{\text{rec}}, \mathbf{c}, \nabla_{\mathbf{c}} \mathcal{L}_{\text{rec}})$ is a learned gradient update term parameterized for a CNN which is trained to minimize the mean squared error between the current iteration $\mathbf{c}^k$ and the ground truth $\mathbf{c}_{gt}$ at each iteration. The fields are then updated:

$$\boldsymbol{\mu}^{k+1} \leftarrow \boldsymbol{\mu}^k - \eta \nabla_{\boldsymbol{\mu}^k} \mathcal{L}_{\text{rec}} \quad (11)$$

$$\mathbf{c}^{k+1} \leftarrow \mathbf{c}^k - \eta \nabla_{\mathbf{c}^k} \mathcal{L}_{\text{rec}} \quad (12)$$

where η is the learning rate.

3 Data Generation

Synthetic data is generated on a discrete domain with a resolution which is a power of two, or which has a small maximum prime factor in each direction to preserve the performance of the Fast Fourier Transform (FFT) used in the pseudo-spectral discrete time (PSDT) simulation method. A perfectly matched layer (PML) is used to absorb waves at the boundary, preventing them from wrapping to the other side of the domain. Visualizing simulations showed that a PML size of 12 on each side of a 128x128 domain was sufficient to prevent wave wrapping and reflection. This margin was subtracted, as well as additional padding for the tissue to avoid generating vessels on the boundary.

The Python script VSystem [6] was used to generate 3D Lindenmayer systems which resemble branching blood vessels (Figure 1). For smaller domains, the system is generated in a larger domain and the image is rescaled using spline interpolation by `scipy.ndimage.zoom` to accommodate the limitations of VSystem.

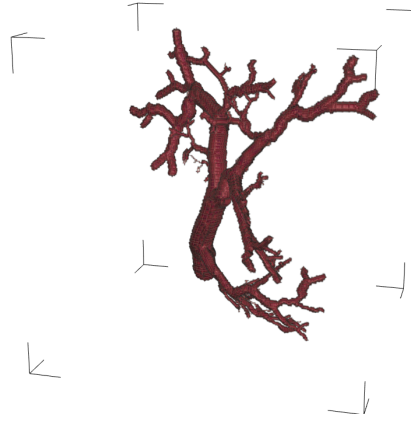


Figure 1: Example of a 3D Lindenmayer system, 207³

For 2D data, the system is then summed along one axis, resulting in the absorption field referred to as μ (Figure 2).

Sensors are generated along a line for 2D data, and in a plane for 3D data within a margin within the PML. These sensors detect the pressure data during the simulation.

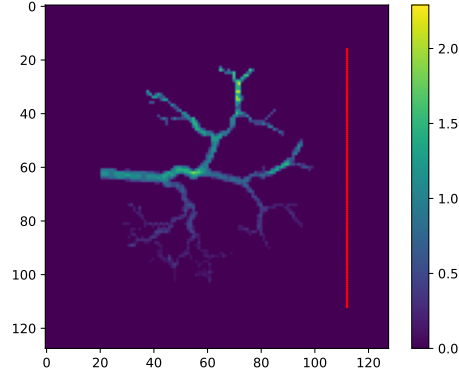


Figure 2: 2D Setup: Absorption field μ and sensors (red)

Attenuation masks (Figure 3) are generated in accordance with Equation 3 by applying an exponential decay to a field of **ones** along a direction for a set of angles. Here, the angles are always evenly spaced (to within the angular resolution of the generated masks) around the domain. There must be sufficient light intensity for a gradient of absorption to exist across the area of interest, so the attenuation coefficient selected for the generation of the masks was such that the gradient was visible. Care was taken to ensure the magnitude of illumination at the center of the image was the same regardless of the angle. The attenuation masks are finally clipped to a value of 1. Figure 3 shows an example with seven attenuation masks applied to μ , giving the set of P_0 , along with the sensor data resulting from the simulations of each field. The strategy of using these illuminations lies in the hypothesis that the visible differences in the P_data will improve the uniqueness of the sound speed field obtained from the reconstruction process.

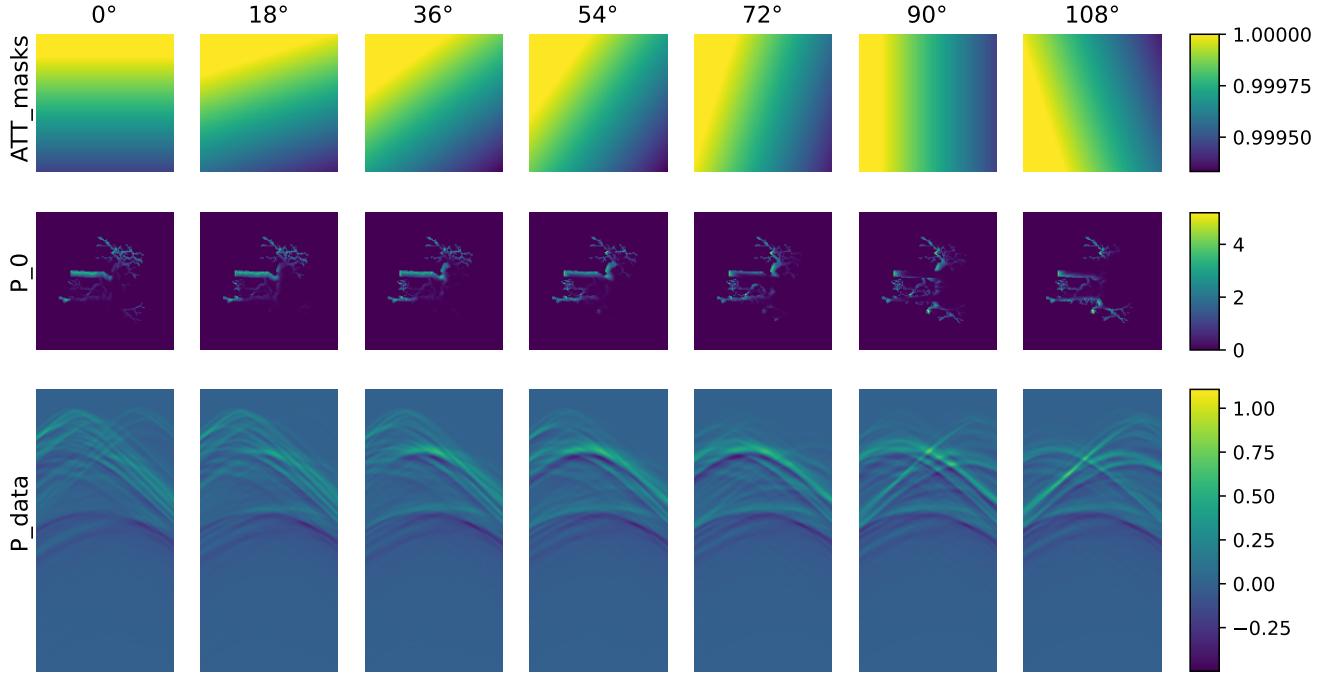


Figure 3: Attenuation masks (top), P_0 (middle) and P_{data} (bottom)

A sound speed field c (Figure 4) is generated using Perlin noise, scaled around the speed of sound in fat: 1450 ± 10 m/s. A mask of the blood vessels μ is added to this field at 1540 m/s to simulate the speed of sound in blood.

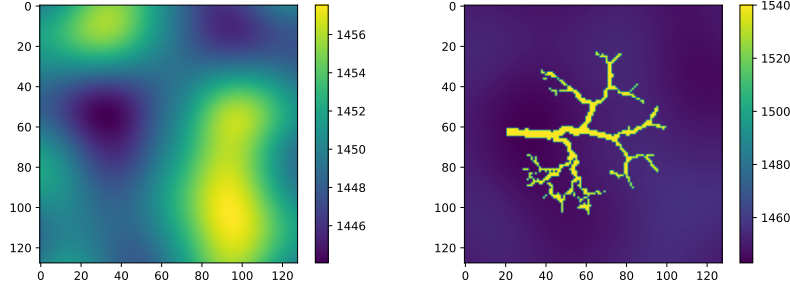


Figure 4: Sound speed field: background noise (left), combined with blood vessels (right)

The forward operator $\mathcal{A}$ is approximated, A , using a PSDT simulation method provided in j-Wave [5], a JAX implementation of k-Wave [7], to generate the data P_data . JAX makes it easy to perform a set of these simulations in parallel on a GPU with the `jax.vmap` function transformation.

Ultrasound transducers are susceptible to noise from various sources, including electronic, thermal and acoustic noise from the environment. Noise in the data diverts a reconstruction from the true solution, it can aggravate solutions with marginal stability and contribute to issues such as getting trapped in local minima early in an iterative reconstruction. The sensors used here are simulated to have unlimited bandwidth and no noise, so Gaussian noise is added to the data to roughly account for the noise in real-world measurements. Gaussian noise is smoothed with a Blackman window as in [5] and added to P_data to improve the realism of the data.

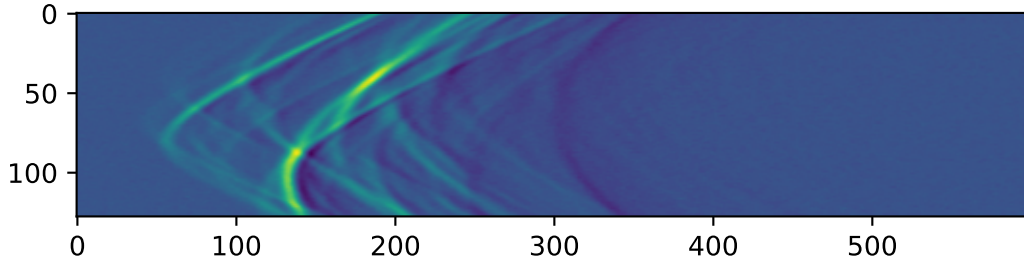


Figure 5: A single p_data_noisy

4 Reconstruction

Algorithm 1 describes the iterative reconstruction process. It uses the function defined in Algorithm 2, which describes a single reconstruction step.

Initializing the reconstruction absorption field μ_r slightly above zero reduces noise in the early iterations of the reconstruction compared to initializing it very near zero. Negative initial pressures are not expected since the region is being illuminated. To naturally enforce non-negativity, the μ_r field is parameterized by the softplus function depicted in Figure 6 and named μ_p . The

softplus function is never equal to zero, but the reconstructed field typically has sufficient contrast even without the use of regularization.

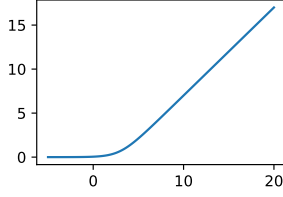


Figure 6: The softplus function $\ln(1 + e^x)$ is used to parameterize the absorption field.

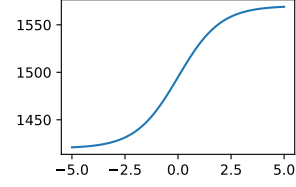


Figure 7: A shifted sigmoid function is used to parameterize the sound speed field.

The reconstruction sound speed field $\mathbf{c_r}$ is initialized to a value low in the bounded range. Parameterization of the sound speed field provides a lower bound which is required to guarantee stability of the simulation and an upper bound based on the expected materials being imaged. Parameterization also serves to normalize the field, significantly improving the optimization performance, particularly given the differing scales of the two fields. The sigmoid function is scaled and centered around the allowable sound speeds (Figure 7) and the field is named $\mathbf{c_p}$. These fields are discretized as Fourier series by the `jaxdf` [4] package as required for simulation and an Adam optimizer is initialized for each field.

The iterative process begins, where the `reconstruction_step` function is called. It returns the summed loss of the data fidelity term and the gradients of the loss with respect to the fields $\mathbf{mu_p}$ and $\mathbf{c_p}$, named $\mathbf{d_mu_p}$ and $\mathbf{d_c_p}$ respectively. The fields are then updated using the gradients and the Adam optimizers. The fields are then de-parameterized for viewing and the process is repeated for the specified number of iterations.

The `reconstruction_step` (Algorithm 2) function defines a loss function `loss_fn` and computes the gradient of the data fidelity loss with respect to the input parameter fields. `loss_fn` receives parameterized fields so backpropagation through `loss_fn` yields gradients of the parameterized versions, $\mathbf{d_mu_p}$ and $\mathbf{d_c_p}$.

These fields are first converted to their on-grid representations for use in the simulator. The field $\mathbf{mu_r}$ is then multiplied by the generated attenuation masks `ATT_masks`, creating the array of initial pressures $\mathbf{P0}$. This happens within `loss_fn` so that a single gradient of the field parameters $\mathbf{d_mu_p}$ is recovered from and the explicit conversion from $\mathbf{P0}$ to $\mathbf{mu_r}$ is avoided. It is assumed that these, or better, attenuation masks could be generated for use in a practical setting.

The batch of data is then simulated using the sound speed field and outputs a batch of predicted sensor pressures $\mathbf{P_pred}$. This is the implementation of the operator $\mathbf{A}$. This batch is subtracted from the noisy data $\mathbf{P_data}$ to obtain the residual.

Reconstructions are performed iteratively by backpropagating the MSE of the data residual through `loss_fn`. This is made easy because the simulator in `j-Wave` is differentiable and compatible with the `jax.grad` function transformation. The field parameters are then updated, using a separate Adam optimizer for each field.

Algorithm 1 Iterative reconstruction

```
1: function RECONSTRUCT(num_illuminations, num_iterations, learning_rates)
2:
3:   mu_r  $\leftarrow$  random.normal(image_shape) ▷ initialize fields
4:   c_r  $\leftarrow$  ones(image_shape) * C + random.normal(image_shape)
5:   c_p  $\leftarrow$  FourierSeries(c_r, domain) / C ▷ discretize and parameterize
6:   mu_p  $\leftarrow$  FourierSeries(mu_r, domain)
7:   ATT_masks  $\leftarrow$  FourierSeries(ATT_masks, domain)
8:
9:   opt_mu  $\leftarrow$  optax.adam(learning_rates[0]) ▷ initialize two Adam optimizers
10:  opt_c  $\leftarrow$  optax.adam(learning_rates[1])
11:
12:  for i in range(num_iterations) do
13:
14:    loss_data, d_mu_p, d_c_p  $\leftarrow$  reconstruction_step(mu_p, c_p,
15:                                                       ATT_masks, P_data, r_mu, r_c) ▷ Algorithm 2
16:    c_p  $\leftarrow$  update_opt_mu(c_p, d_c_p) ▷ update parameter fields
17:    mu_p  $\leftarrow$  update_opt_c(mu_p, d_mu_p)
18:
19:    mu_r  $\leftarrow$  get_mu(mu_p) ▷ de-parameterize with softplus
20:    c_r  $\leftarrow$  get_c(c_p) ▷ de-parameterize with sigmoid
21:  return loss_data, mu_r, c_r
```

Algorithm 2 Reconstruction step

```
1: function RECONSTRUCTION_STEP(mu_p, c_p, ATT_masks, P_data, r_mu, r_c)
2:
3:   function LOSS_REC(mu_p, c_p)
4:
5:     mu_r  $\leftarrow$  get_mu(mu_p)
6:     c_r  $\leftarrow$  get_c(c_p) ▷ de-parameterize fields
7:     P0  $\leftarrow$  mu_r * ATT_masks ▷ create initial pressures
8:     medium  $\leftarrow$  Medium(domain, sound_speed=c_r) ▷ re-initialize simulation medium
9:     P_pred  $\leftarrow$  A(medium, time_axis, P0) ▷ simulate wave propagation
10:
11:     return mse(P_pred, P_data) ▷ Sum of mean-squared error of residual
12:
13:   value_and_grad_fn  $\leftarrow$  value_and_grad(loss_rec, argnums=(0, 1))
14:   ▷ JAX syntax to transform loss function to one which computes the gradient of the returned
    loss with respect to arguments (argnums) numbered 0: mu_p and 1: c_p .
15:
16:   loss_data, (d_mu_p, d_c_p)  $\leftarrow$  value_and_grad_fn(mu_p, c_p)
17:   ▷ compute the loss and gradients of the parameters
18:   return loss_data, d_mu_p, d_c_p
```

Figure 8 depicts a reconstruction using the algorithm above with 4 illuminations and a single Adam optimizer. The recovery of μ plateaus and c assumes an arbitrary field which allows deep minimization of the data loss

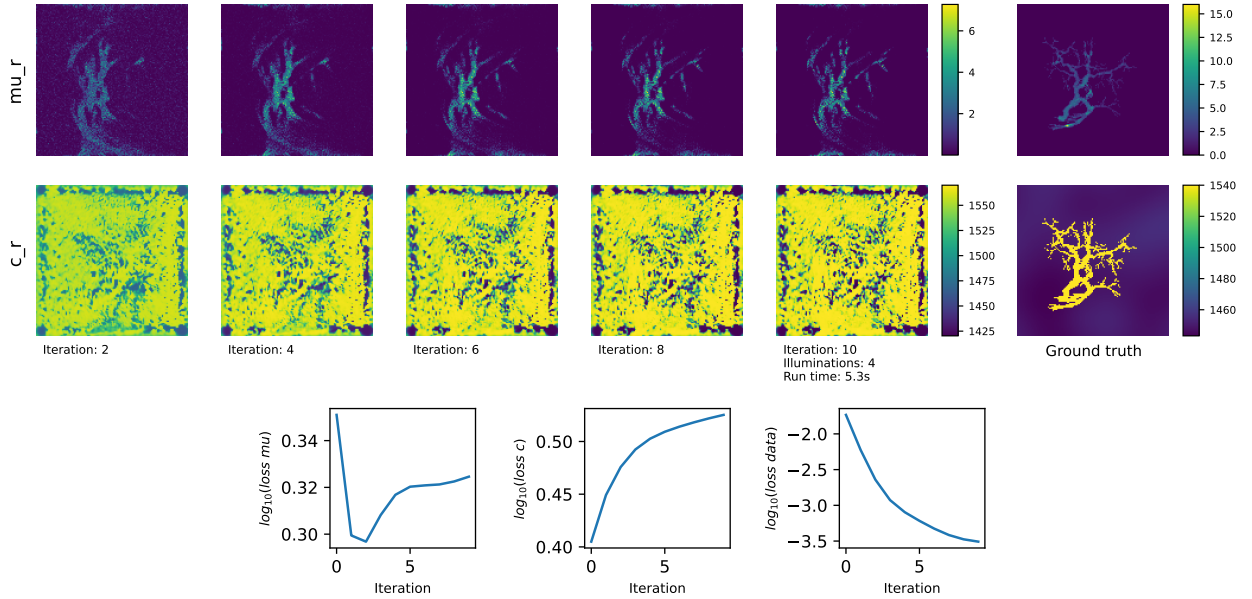


Figure 8: Reconstruction using Algorithm 1 with a single optimizer

Figure 9 again shows a reconstruction using the same algorithm with the exception that the gradient for c is replaced by the gradient for μ in every iteration. In fact, supplying the c gradient or a combination of the two at any iteration seems only to degrade the reconstruction. By taking advantage of the parameterization and initializations of the reconstruction, this approach leverages the structural similarity and directly translates the structure of the absorption field to the sound speed field.

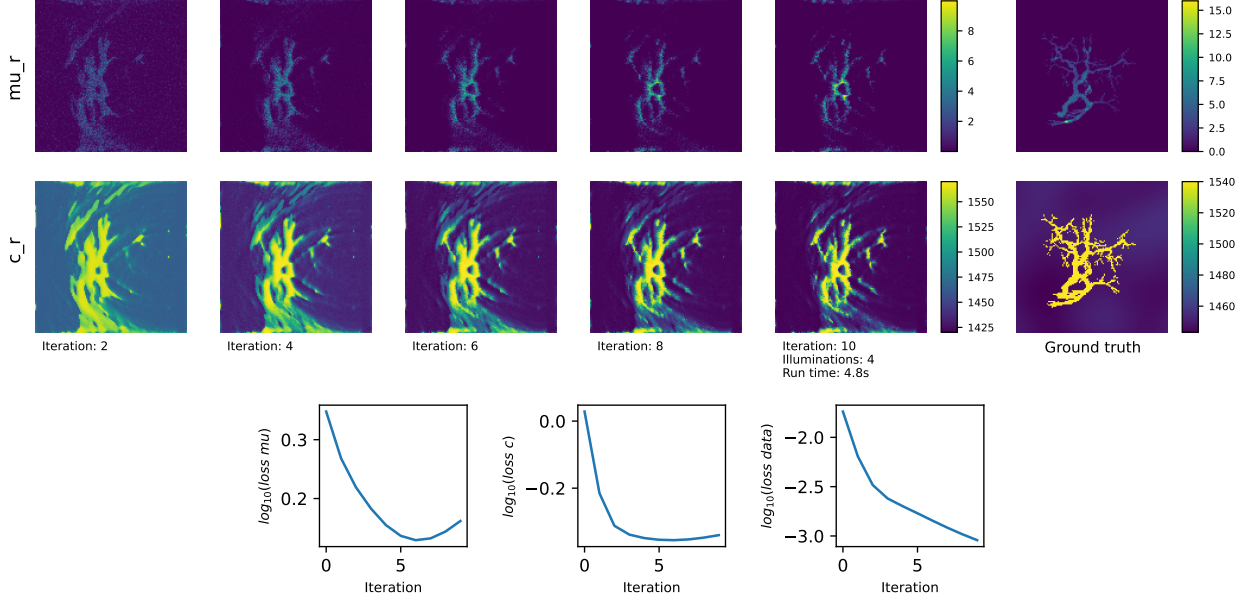


Figure 9: Reconstruction feeding the gradient of μ to c

Using two separate Adam optimizers, the reconstruction was attempted using a high number of iterations and no gradient-sharing. Figure 10 shows the result. There is only a faint hope of structure.

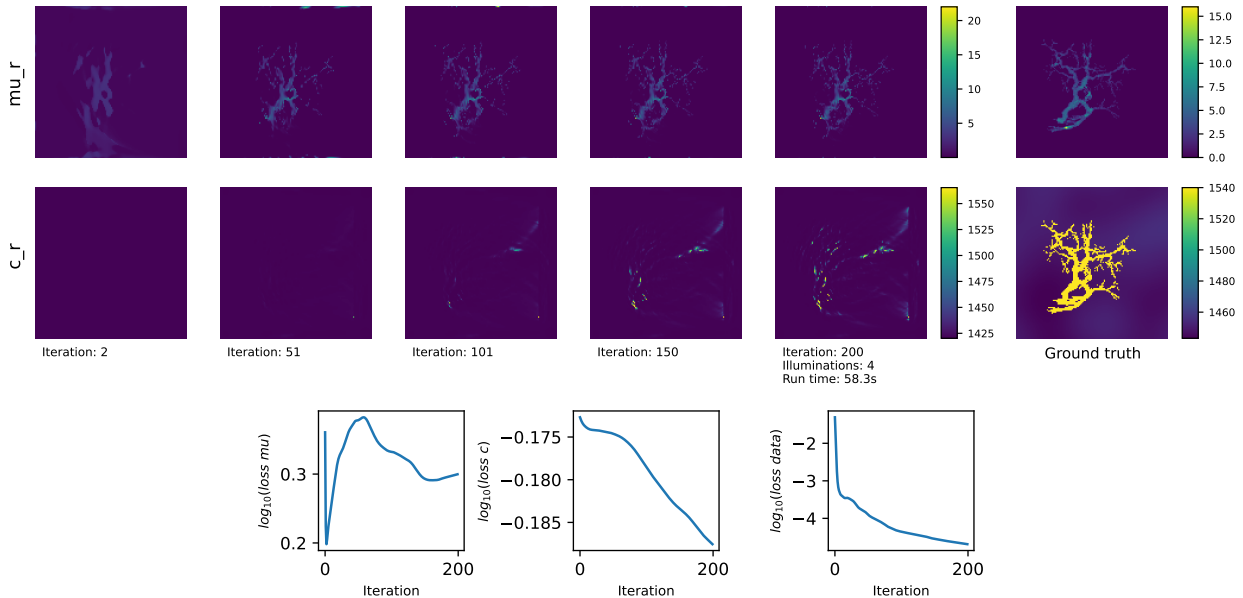


Figure 10: Reconstruction using two Adam optimizers without gradient-sharing

Finally, the unregularized reconstruction with gradient-sharing using two separate Adam optimizers is shown in Figure 11. The separate learning rates prove useful. Note that the background Perlin noise of the sound speed field is seemingly unrecoverable. Even when updating with the gradient of the sound speed field on a schedule, the c field loss plateaus. An optimization technique based on spatial content will be necessary to recover this feature due to its subtle pattern and differing scale. Failure to recover this feature would likely lead to features in the fields being smoothly shifted to the wrong position. The current reconstruction method relies on this misalignment being translated to low frequency background noise which is an unlikely outcome without explicit regularization. Mathematically formulating this relationship is challenging, but a learned regularizer could potentially be trained to recover this translation.

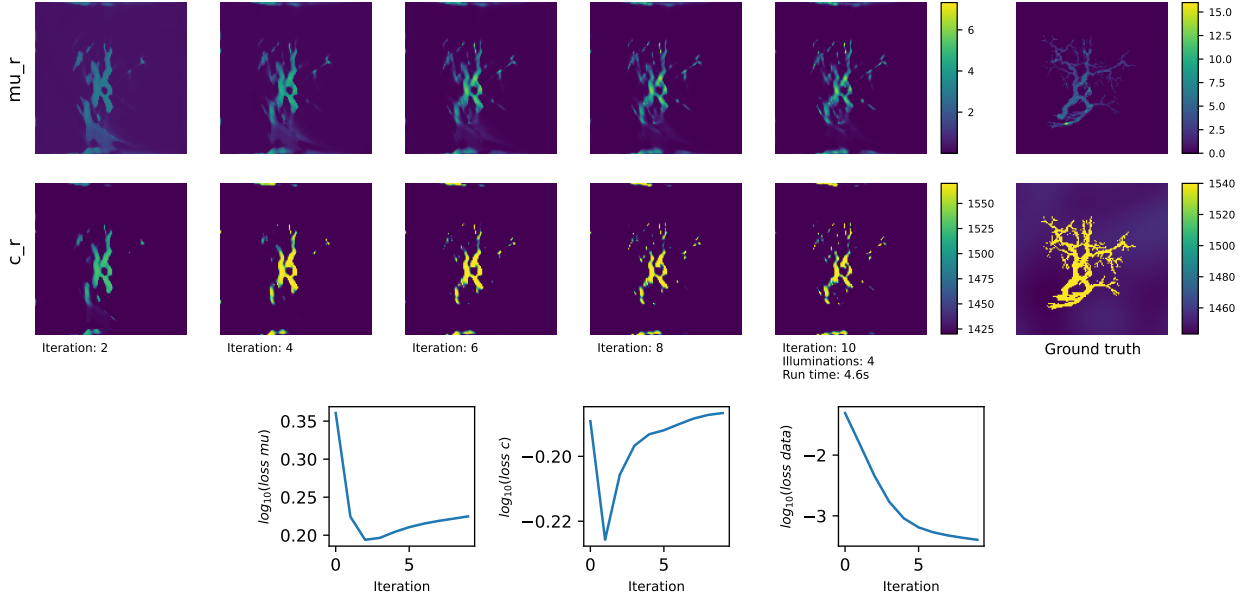


Figure 11: Reconstruction using two Adam optimizers without gradient-sharing

While representing fundamentally different physical influences, the sound speed and attenuation fields often possess a structural similarity. Without sharing information between them, reconstructions result in comparatively homogeneous sound speed fields which result in recognizable structures in the absorption field and simply minimize the data loss. Using multiple illuminations alone is not enough to obtain significant structure in the sound speed field. Also, simply sharing the gradient in this way may not be a general solution to regularizing the sound speed field. This solution feels manufactured because with the use of generated data, the relationship between the sound speed and absorption fields is clearly known. If the relationship between the structures is not this simple, which it may not be in practice, this method may not be effective. If nothing else, this experiment accentuates an avenue in the problem of regularizing the sound speed field: learning the similarity between these fields.

4.1 Multiple Illuminations

To improve the reconstructed sound speed field the use of multiple illumination angles is considered. The reconstructions in this section use the gradient-sharing strategy employed in the previous section.

As in Figure 3, an illumination of zero degrees correspond to emanation from the top edge of the domain. The illumination angles are evenly spaced around the domain to within the radial resolution of the generated data and measurements would be taken one at a time. Figure 12 depicts unregularized reconstructions of noiseless data completed with varying numbers of illuminations and approximately the same computation time. In the given computation time, the data loss is reduced most effectively by a single illumination, yet the reconstruction losses remain the highest. This overfitting highlights the non-uniqueness of the solution. With even two angles, the reconstruction converges to a better solution with much of the shadow of noise at the bottom of the image eliminated.

For this particular setup, between 3 and 5 illuminations produced the best reconstructions. The overlapping loss curves in Figure 13 suggest that the additional illuminations give diminishing benefits. While additional illuminations eliminate localized smear patterns in the reconstructions, sequential iterations are needed to improve the overall reconstruction. Also note that the μ loss continues to decline at high iterations, but the c loss still plateaus and begins to overfit, albeit at a lower level.

The use of multiple illuminations significantly reduces loss in the fields in fewer iterations (Figure 13). Additionally, the average computation time per simulation decreases rapidly with the inclusion of the first several illuminations (Figure 15). In contrast, additional iterations with a single illumination must be executed sequentially, which may leave GPU resources underutilized.

A key advantage of employing multiple illuminations is the ability to perform simulations in parallel, thereby maximizing the GPU's computational capacity. However, parallel and serial reconstructions are not equivalent. Reconstructions with a high number of illuminations tend to make less overall progress when the attempted parallelization exceeds the GPU's compute capacity because time is then spent calculating different illuminations of the same iteration rather than the next iteration. The results may vary depending on the GPU's compute capacity and RAM availability. In a larger 3D domain, significant (but feasible) GPU capacity would be required to perform simulations in parallel and take advantage of the potential time-saving benefits associated with using multiple illuminations.

Using illuminations from multiple angles yields a noticeable improvement in reconstruction quality and improves the uniqueness of the sound speed field solution.

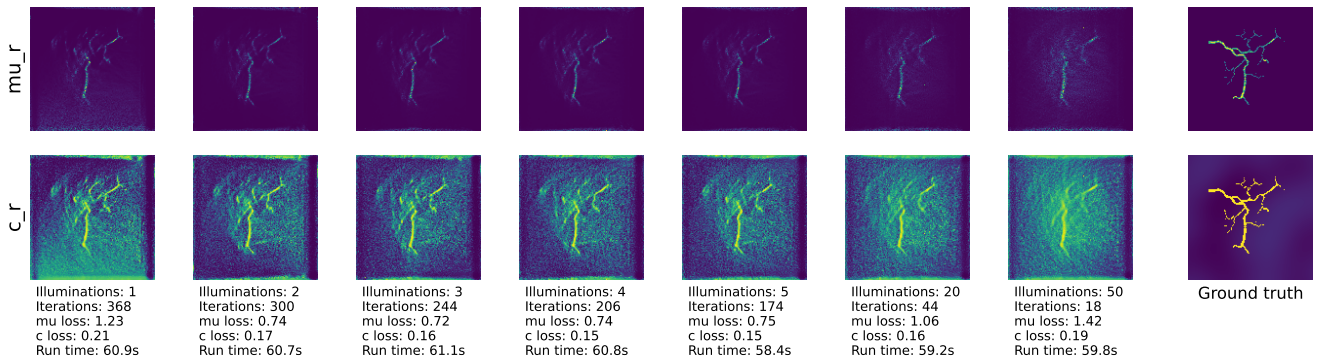


Figure 12: Reconstructions with varying numbers of illuminations with a fixed computation time.

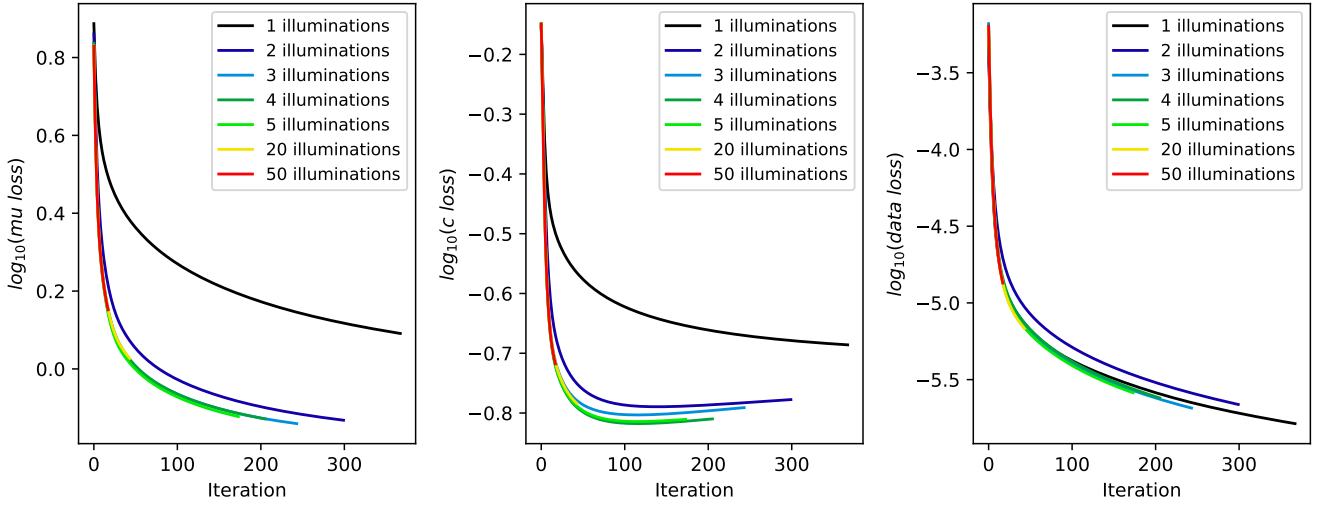


Figure 13: Reconstruction losses with multiple illuminations over iterations with a fixed computation time.

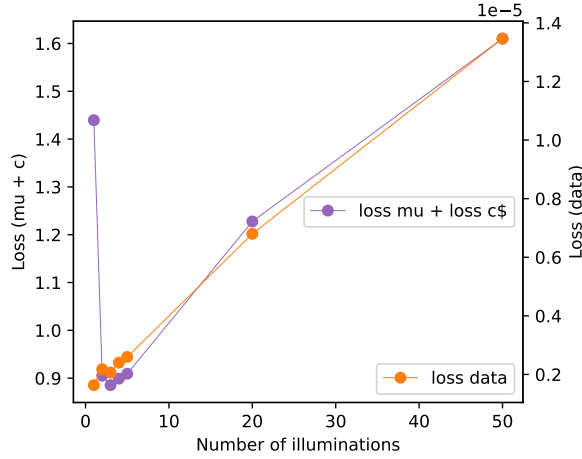


Figure 14: Losses by number of illuminations with a fixed computation time.

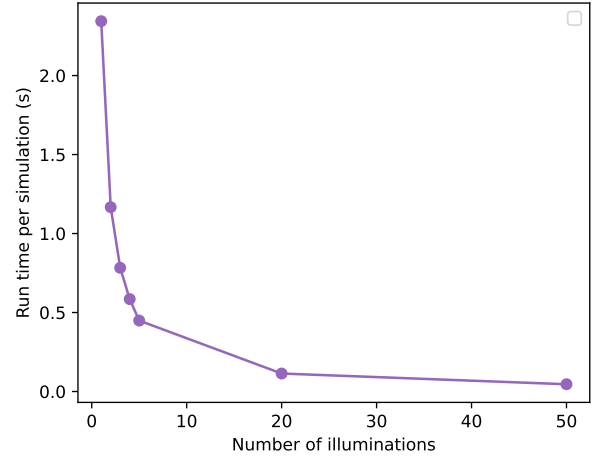


Figure 15: Average run time per simulation by number of illuminations

4.2 Regularization

As discussed, changes in the spatial amplitudes within the chosen sound speed field result in translations of the structure in the absorption field. A mathematical relationship of this translation is difficult to formulate, but a learned regularizer could potentially recover this translation and other relationships between the fields.

In the implementation used here, the gradient of the sound speed field parameters is modified by a CNN before it is used to update the field. This allows the continued use of the Adam optimizer for updating the field parameters, which is shown in Appendix A to offer a significant benefit over a simple gradient descent. Intuitively, a momentum-based optimizer should be well-suited for updating the parameters of two fields which are interrelated, among other things, by spatial

amplitude and translation.

The reconstructions in Figures 16 and 17 depicts the current capability of the regularizer. The network (called TreeNet in the source code) takes field parameters μ_p , c and field gradients from backpropagating the data fidelity loss, d_{μ_p} , d_{c_p} . The network encodes each field separately, concatenates the fields, convolves, then decodes to a single field using skip connections from all other fields. The implementation of this network clearly needs additional work, but the key takeaway is that the network is capable of learning a relationship between the fields using the sound speed reconstruction and gradients. Reconstructing the background Perlin noise is within closer reach. Reconstructing through materials with high sound speed such as bone may also benefit from such a regularizer.

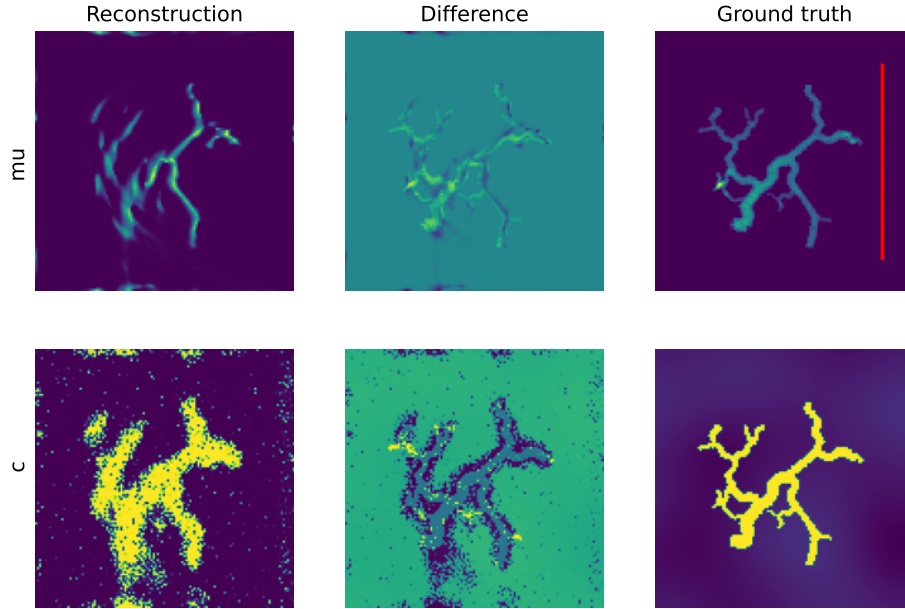


Figure 16: Reconstruction using a learned regularizer for the sound speed parameter gradients

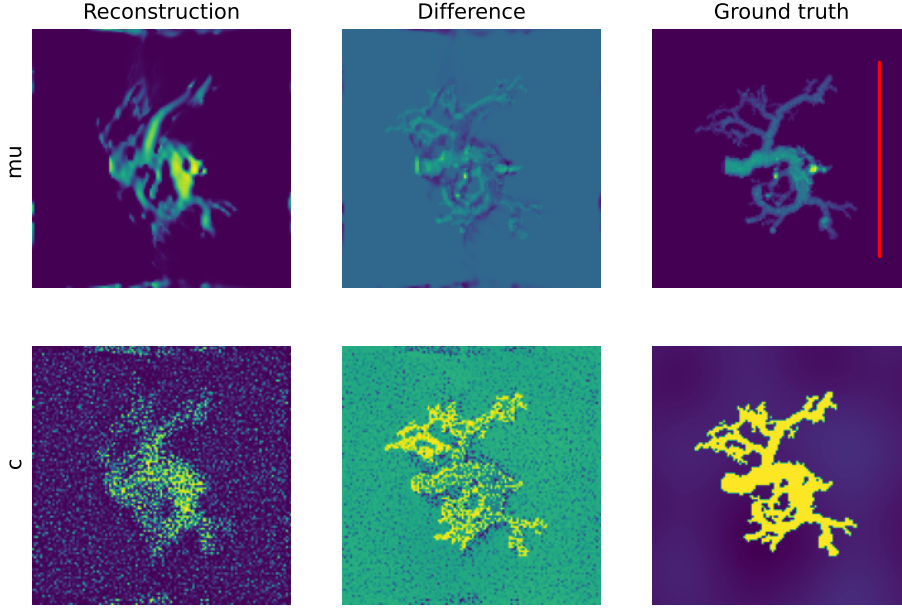


Figure 17: Reconstruction using a learned regularizer for the sound speed parameter gradients

The training loop implementation exposes countless options for experimentation, so the current state of the program is elucidated. The forward model is computed separately from the training loop, saving potentially significant computational effort backpropagating losses through the simulator. The update for μ is stable, and proven to be beneficial to the c reconstruction, so the μ update is performed prior to running the gradient transforming network. This training process suggests a small update to Equation 10. Let the gradient of the data loss with respect to c be $d_c = \nabla_{\mathbf{c}} \|\mathbf{P}_{\text{data}} - \mathcal{A}(\mathbf{P}_0, \mathbf{c})\|_2^2$ and $d_\mu = \nabla_{\mu} \mathcal{L}_{\text{rec}}$, which could optionally include some regularization of its own, but it doesn't here. The updated μ^{k+1} is used in the network $\mathcal{R}$, while $\mathbf{c}$ awaits the transformed d_c before undergoing its update:

$$\tilde{d}_{\mathbf{c}}^k = \mathcal{R}_{\mathbf{c}}(\mu^{k+1}, d_{\mu}^k, \mathbf{c}^k, d_{\mathbf{c}}^k) \quad (13)$$

$$\mathbf{c}^{k+1} \leftarrow \text{Adam}(\mathbf{c}^k, \tilde{d}_{\mathbf{c}}^k) \quad (14)$$

In Algorithm 3, Adam optimizers are initialized for each of the parameter fields. In a loop, the reconstruction step from Algorithm 2 is called to obtain gradients for the field parameters with respect to the data loss and the parameters μ_{p} are updated. A loss function is defined which applies the model to the fields and their gradients to obtain the modified c gradient which is then used to update the parameters. The field is deparameterized so the MSE loss can be computed. The loss function is transformed to a function which computes the gradient of the loss with respect to the model parameters. The loss and gradients are then computed and the model parameters are updated. The process is repeated for the specified number of iterations.

Inspiration for this model was taken from [10], where a regularizer was learned for a single p_0 reconstruction using a similar method of encoding the field and its gradient separately, concatenating and decoding to a single image using skip connection. Note the use of indexing on the model parameters $\mathbf{r}_{\text{p}}$. A different set of weights is used for each iteration. This is required with this architecture to achieve any desirable effects.

An interesting modification involves providing the model with the parameterized set of initial pressures $P0_p$ rather than the field μ_p . In theory, this would further deepen the benefits of using multiple illumination angles, and potentially give a well-structured network the information required to carry out a clean reconstruction. The results in Figure 18 show that significant work remains, but the reconstructed sound speed field contains structure and a more smooth variation of the background region.

An alternative method involves learning the next step in the iterative reconstruction, foregoing the field parameter optimizer. Speculatively, doing so essentially requires the network to learn a suitable stepping process for the field, which may exhibit the physical connection discussed. This could be done using a network which has a suitable inductive bias to perform the translation across iterations of the reconstruction. Using different network parameters for each iteration of the reconstruction as is done here could potentially discretize a suitable process. However, incorporating momentum into this method seems valuable, as it may better account for the dynamics between the fields. This method is worth further investigation.

It is justifiable to use regularization on both fields. At this stage, the reconstruction of μ is well-behaved, and attempts at regularizing it destabilized attempts at regularizing c .

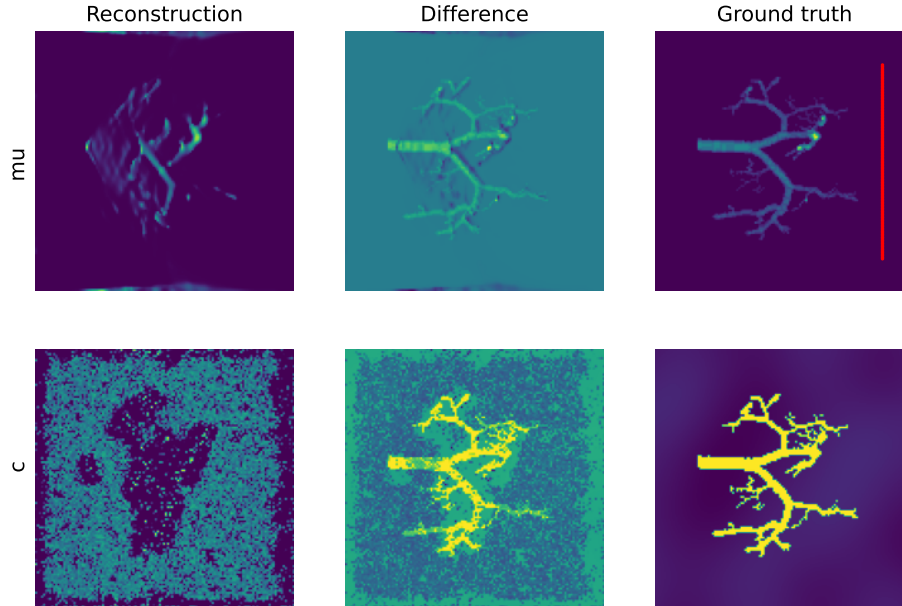


Figure 18: Reconstruction using gradient sharing and a network trained on the initial pressures from four illuminations

Algorithm 3 Learning Sound Speed Regularization

```
1: function TRAIN_R_C(num_iterations, learning_rates)
2:
3:   ATT_masks  $\leftarrow$  FourierSeries(data["ATT_masks"], domain)     $\triangleright$  Initialize and discretize
4:   c_p  $\leftarrow$  FourierSeries(zeros(image_shape) - 5., domain)
5:   mu_p  $\leftarrow$  FourierSeries(zeros(image_shape) + 1, domain)
6:   opt_mu  $\leftarrow$  optax.adam(learning_rate_mu)
7:   opt_c  $\leftarrow$  optax.adam(learning_rate_c)
8:
9:   for i in range(num_iterations) do
10:                                      $\triangleright$  Run reconstruction step and get gradients
11:     loss_data, d_mu_p, d_c_p  $\leftarrow$  reconstruction_step(mu_p, c_p,
12:                                                         ATT_masks, P_data)
13:
14:     mu_p  $\leftarrow$  update_opt(mu_p, d_mu_p)     $\triangleright$  Update mu_p with optimizer
15:     mu_r  $\leftarrow$  get_mu(mu_p)                 $\triangleright$  Deparameterize
16:
17:     function LOSS_FN(r_p, mu_p, d_mu_p, c_p, d_c_p)
18:                                      $\triangleright$  Apply model using parameters r_p to obtain modified gradient
19:     d_c_p, updates  $\leftarrow$  model[i](r_p, mu_p, d_mu_p, c_p, d_c_p)
20:
21:     c_p  $\leftarrow$  update_opt(c_p, d_c_p)     $\triangleright$  Apply updates to c_p
22:     c_r  $\leftarrow$  get_sound_speed(c_p)     $\triangleright$  Deparameterize
23:     loss_c  $\leftarrow$  mse(c_r, c)     $\triangleright$  Compute the loss for c_r
24:
25:     return loss_r
26:
27:      $\triangleright$  Transform loss_fn to gradient of loss_c with respect to model parameters r_p
28:     value_and_grad_fn  $\leftarrow$  value_and_grad(loss_fn, argnums=(0), has_aux=True)
29:
30:                                      $\triangleright$  Compute the loss and gradients
31:     loss_r, d_r_p  $\leftarrow$  value_and_grad_fn(r_p[i], mu_p, d_mu_p, c_p,
32:                                             d_c_p, ATT_masks)
33:     r_p[i]  $\leftarrow$  r_p[i].apply_gradients(grads=d_r_p)     $\triangleright$  Apply the gradients to the model
34:     parameters
```

5 Further Work

5.1 Arbitrary sensor positions

The workflow used here preserves considerable flexibility regarding arbitrary sensor positions. In theory, a flexible transducer array which bends to the contours of the subject could be used, given sufficiently accurate estimation of the sensor positions attainable perhaps using bend sensors. Such devices have been developed, though work remains to be done to expand their bandwidth to the range required to achieve the full benefits of photoacoustic imaging [3]. Incorporating bandwidth limitations and a different array of sensor positions for each successive measurement is particularly easy using j-Wave.

5.2 A more complete illumination model

A simplifying assumption was made in Equation 3 which leads to the treatment of illumination as multiplication with a mask. This eliminates much of the nuance of a real light field such as shadowing effects which help in localizing complex features at different depths. Without making this assumption, the light must be tracked as it transits the domain, leading to a computational burden in data generation and backpropagating residuals. Speculatively, much of the benefit of using multiple illuminations lies in this transformation. Generating data in accordance with the more complete Equation 2, then performing reconstructions which update the attenuation field as well may accomplish much of what the gradient-transforming CNN is attempting to approximate.

6 Conclusion

The methods explored here focus on the ill-posed problem of simultaneously recovering absorption and sound speed fields in photoacoustic imaging. A flexible process for generating 2D and 3D data was developed with a simplified illumination model. The reconstruction process was found to benefit from parameterization and the use of Adam optimizers for updating field parameters. Gradient sharing was identified as a key component of obtaining structure in the sound speed field. The use of multiple illumination angles was found to reduce certain artifacts, enable parallelization in the reconstruction, and generally improve reconstruction quality. In an effort to extend the variety of information shared between the fields and the performance on multiple scales, a convolutional neural network which encoded both fields and their gradients was used to transform the sound speed field parameter gradients. Further work should focus on utilizing a more complete illumination model and identifying a network architecture with suitable characteristics to regularize the sound speed field.

A Reconstructions

These reconstructions involve poor decisions, but are included to highlight the importance of optimization methods, prior information, normalization and bounding. In particular, the following reconstructions are done without parameterizing either field or by working with the Fourier series discretizations of the fields, but by working with the fields directly. To recap, the fields are on different scales, with μ approximately in the range $[0, 20)$ and c in the range $[1430, 1550]$. Without normalizing the sound speed, the optimization problem is ill-conditioned.

These fields are updated using simple gradient descent steps with separate learning rates. Adjusting the learning rates to achieve even these poor results was tedious.

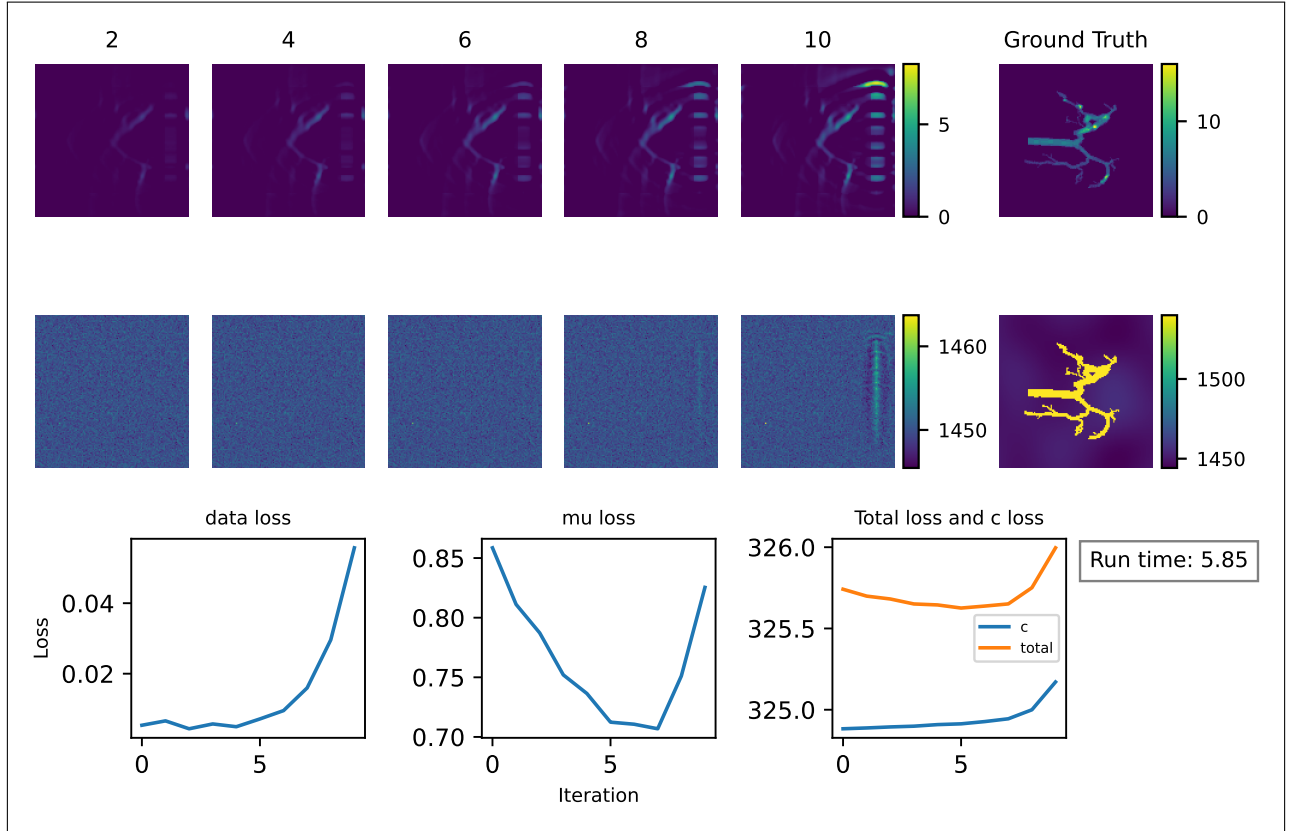


Figure 19: Simple gradient descent

Updating fields using an Adam optimizer moved the process in a positive direction. Structure appears in the sound speed field, but the reconstruction still diverges, exacerbated by the fact that neither field has been normalized in any way.

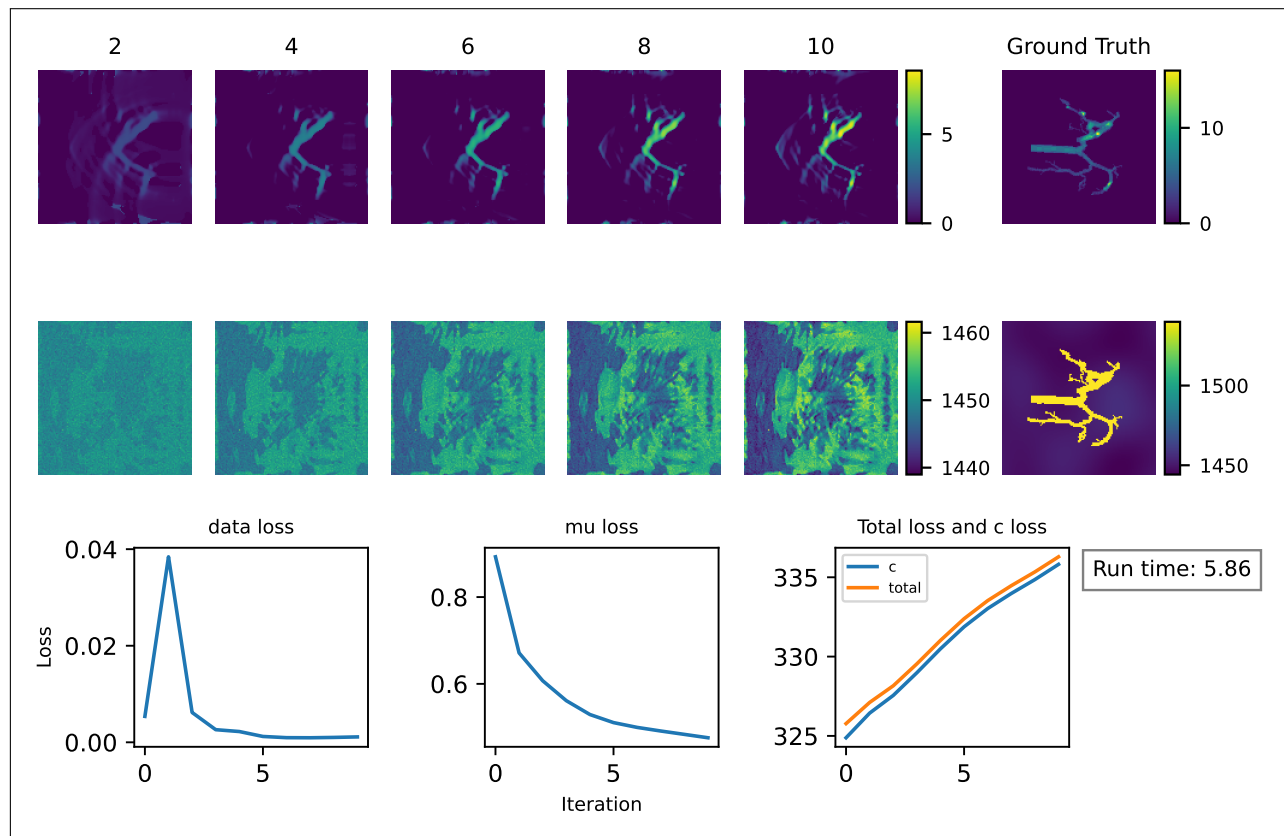


Figure 20: Fields updated with a single Adam optimizer

This is an important result, as it was the first sign of useful structure in the sound speed field. Sharing the gradient from the absorption field with the sound speed field gives the sound speed field enough prior information to partially converge. The reconstruction still suffers from the lack of normalization and the sound speed field is not able to attain the correct scale.

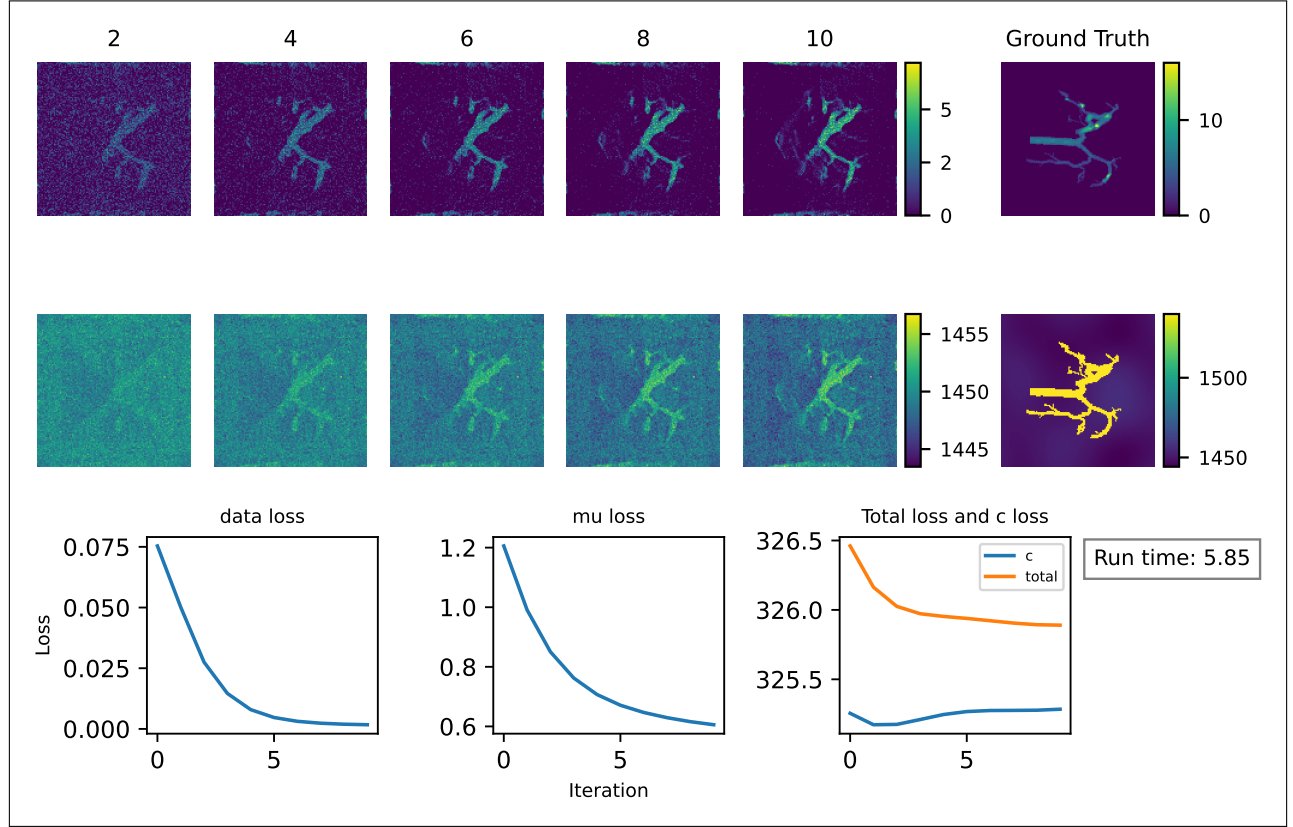


Figure 21: Fields updated with two Adam optimizers

Normalizing μ and c by subtracting the mean and dividing by the standard deviations of the ground truth fields was meant to improve the conditioning and the sound speed field, and it is able to attain a more appropriate scale. Approximate values, particularly for the sound speed field must be known, which is a realistic assumption. This reconstruction still diverges because appropriate upper and lower bounds were not imposed on the sound speed field at this stage. Parameterization of the fields takes care of normalization and bounding.

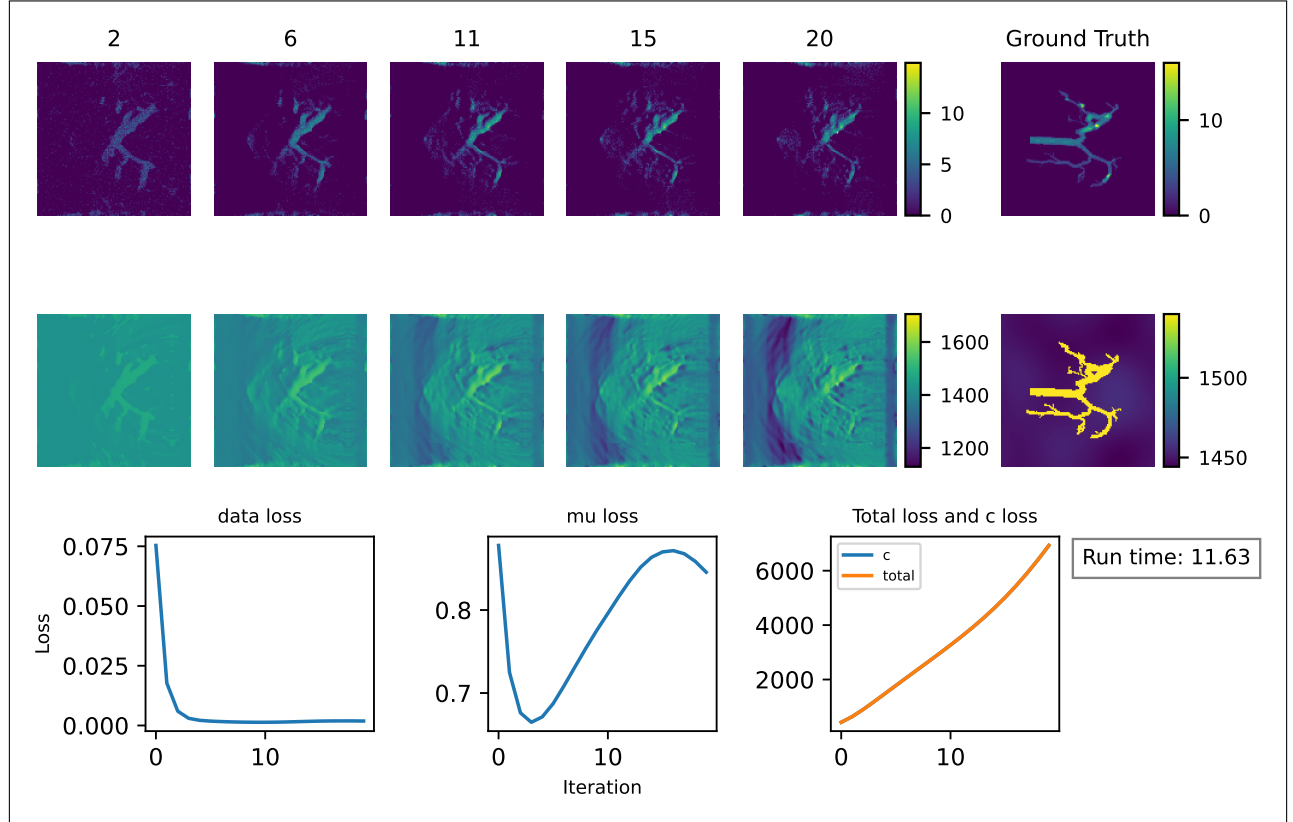


Figure 22: Normalized fields updated with two Adam optimizers. No bounds imposed on sound speed

References

- [1] Paul Beard. Biomedical photoacoustic imaging. *Interface Focus*, 1(4):602–631, August 2011.
- [2] Mithat Idemen. Some Universal Properties of the Green’s Functions Associated with the Wave Equation in Bounded Partially-Homogeneous Domains and Their Use in Acoustic Tomography. *Applied Mathematics*, 08(04):483–499, 2017.
- [3] Thanh-Giang La and Lawrence H. Le. Flexible and Wearable Ultrasound Device for Medical Applications: A Review on Materials, Structural Designs, and Current Challenges. *Advanced Materials Technologies*, 7(3):2100798, 2022.
- [4] Antonio Stanziola, Simon Arridge, Ben T. Cox, and Bradley E. Treeby. A research framework for writing differentiable PDE discretizations in JAX. *Differentiable Programming workshop at Neural Information Processing Systems 2021*, 2021.
- [5] Antonio Stanziola, Simon R. Arridge, Ben T. Cox, and Bradley E. Treeby. J-Wave: An open-source differentiable wave simulator. 2022.
- [6] Paul W. Sweeney, Lina Hacker, Thierry L. Lefebvre, Emma L. Brown, Janek Gröhl, and Sarah E. Bohndiek. Unsupervised segmentation of 3D microvascular photoacoustic images using deep generative learning, February 2024.
- [7] Authored Bradley Treeby, Ben Cox, and Jiri Jaros. A MATLAB toolbox for the time domain simulation of acoustic wave fields User Manual.
- [8] Bradley E. Treeby and B. T. Cox. Modeling power law absorption and dispersion for acoustic propagation using the fractional Laplacian. *The Journal of the Acoustical Society of America*, 127(5):2741–2748, May 2010.
- [9] Lihong V. Wang and Song Hu. Photoacoustic Tomography: In Vivo Imaging from Organelles to Organs. *Science (New York, N.y.)*, 335(6075):1458–1462, March 2012.
- [10] Tong Wang, Menghui He, Kang Shen, Wen Liu, and Chao Tian. Learned regularization for image reconstruction in sparse-view photoacoustic tomography. *Biomedical Optics Express*, 13(11):5721–5737, October 2022.
- [11] Yong Zhou, Junjie Yao, and Lihong V. Wang. Tutorial on photoacoustic tomography. *Journal of Biomedical Optics*, 21(6):061007, June 2016.